

Advanced (Carolina Reaper)

- Q1.** A bakery sells x loaves of bread at

$$2$$

each. They also have a discount of

$$5$$

for every 5 loaves sold. The revenue R from selling x loaves is given by $R = 2x - 5\left\lfloor \frac{x}{5} \right\rfloor$. What is the quadratic equation that models the revenue when x is between 10 and 15 loaves, and x must be a whole number?

- (A) $R = 2x - 5$
 (B) $R = 2x - 5x^2$
 (C) $R = x^2 + 2x - 5$
 (D) $R = -x^2 + 2x - 5$
 (E) $R = x^2 - 2x + 5$
- Q2.** What is the discriminant of the quadratic equation $x^2 + 4x + 4 = 0$?
- (A) 0
 (B) 4
 (C) 8
 (D) 12
 (E) 16
- Q3.** A group of friends want to go on a road trip that is 240 miles long. If they drive at an average speed of x miles per hour, the time it takes to complete the trip is given by $t = \frac{240}{x}$. However, they also have to account for rest stops, which add an extra $2x^2$ hours to the trip. What is the quadratic equation that models the total time of the trip?
- (A) $t = \frac{240}{x} + 2x^2$
 (B) $t = \frac{240}{x} - 2x^2$
 (C) $t = 2x^2 + \frac{240}{x}$
 (D) $t = 2x^2 - \frac{240}{x}$
 (E) $t = \frac{240}{x} + x^2$
- Q4.** What is the nature of the solutions of the quadratic equation $x^2 + 2x + 5 = 0$?
- (A) Two distinct real solutions
 (B) One repeated real solution
 (C) Two complex conjugate solutions
 (D) One complex solution
 (E) No real solutions

- Q5.** A water tank can hold 1000 gallons of water. The amount of water in the tank at any given time t (in hours) is given by $V = 1000 - 2t^2$. What is the quadratic equation that models the volume of water in the tank?

- (A) $V = 1000 - 2t^2$
 (B) $V = 2t^2 - 1000$
 (C) $V = 1000 + 2t^2$
 (D) $V = -2t^2 + 1000$
 (E) $V = t^2 - 1000$

- Q6.** A company produces x units of a product at a cost of

$$5$$

per unit. They also have a fixed cost of

$$200$$

. The total cost C of producing x units is given by $C = 5x + 200$. However, they also have to account for a loss of x^2 units due to defects. What is the quadratic equation that models the total cost of producing x units?

- (A) $C = 5x + 200 - x^2$
 (B) $C = 5x + 200 + x^2$
 (C) $C = 200 - 5x - x^2$
 (D) $C = 200 + 5x - x^2$
 (E) $C = x^2 + 5x + 200$

- Q7.** A cyclist travels at an average speed of x miles per hour. The time it takes to complete a 20-mile trip is given by $t = \frac{20}{x}$. However, they also have to account for a headwind that reduces their speed by $0.5x^2$ miles per hour. What is the quadratic equation that models the time it takes to complete the trip?

- (A) $t = \frac{20}{x} + 0.5x^2$
 (B) $t = \frac{20}{x} - 0.5x^2$
 (C) $t = 0.5x^2 + \frac{20}{x}$
 (D) $t = 0.5x^2 - \frac{20}{x}$
 (E) $t = \frac{20}{x} + x^2$

- Q8.** A quadratic equation has a discriminant of -16 . What can be said about the nature of its solutions?

- (A) Two distinct real solutions
 (B) One repeated real solution
 (C) Two complex conjugate solutions
 (D) One complex solution
 (E) No real solutions

Open Ended Questions (FRQ)

- Q9.** A farmer has 240 feet of fencing to enclose a rectangular area. If the length of the rectangle is x feet, the width is $\frac{240-2x}{2}$ feet. Write a quadratic equation that models the area of the rectangle, and determine the nature of its solutions.
- Q10.** A projectile is launched from the ground with an initial velocity of 20 feet per second. The height h of the projectile at any given time t (in seconds) is given by $h = -16t^2 + 20t$. Write a quadratic equation that models the height of the projectile, and determine the time at which the projectile reaches its maximum height.

Chef's Tips

- Read each question carefully before answering
- Use the quadratic formula to solve quadratic equations
- Check your work carefully to avoid errors